[Paper Title]

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Abstract

1 Introduction

Contributions.

- TODO: contribution 1.
- TODO: contribution 2.
- TODO: contribution 3.

2 Related Work

TODO: subtopic 1.

TODO: subtopic 2.

TODO: subtopic 3.

3 Method

3.1 Preliminaries and Problem Setup

3.2 Model Architecture

To enable history-dependent reasoning, we augment the policy with an explicit memory state $\mathcal{M}_t$:

$$a_t \;\sim\; \pi(\cdot \mid l,\; o_t,\; \mathcal{M}_t)\,. \tag{1}$$

Memory token block. We expose the memory sequence $\mathcal{M}$ produced in Sec. 3.3 to the backbone as a single token block $\mathcal{M}_t \in \mathbb{R}^{B\times d}$ with fixed budget B (zero-padded when fewer tokens are available). The two-token-kind structure of $\mathcal{M}$ — TokenDropped keyframe patches and MotionJEPA motion tokens — is opaque to the backbone: from the policy's perspective $\mathcal{M}_t$ is just a length-B sequence of d-dimensional vectors carrying time tags.

3.3 Memory Token Generation and Training

The memory passed to the VLA is a sparse, time-aligned sequence of two complementary token kinds drawn from a single robot video of length T: (i) *keyframe tokens*, which anchor the scene at semantic transitions, and (ii) *motion tokens*, fixed-stride summaries of short-range visual change produced by a JEPA-style encoder. Both kinds are computed offline from the raw video and a synchronized proprioceptive trace; no gradient flows back into the perception backbones during VLA training.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Figure 1: TODO: method overview figure. Brief description of the figure plus the key take-away message.

Notation. Let $\{I_t\}_{t=0}^{T-1}$ be the input frames and $\{s_t\}$ the proprioceptive state. We use a frozen vision tokenizer $\Phi_{\text{SigLIP}} : \mathbb{R}^{3\times H\times W} \to \mathbb{R}^{S\times E}$ with $S = 256$ patches and $E = 1152$. A separate frozen image–text backbone Φ_{CLIP} is used only as a scalar similarity function for keyframe detection. We write $x_t = \Phi_{\text{SigLIP}}(I_t) \in \mathbb{R}^{S\times E}$.

Keyframe detection. Keyframes mark moments when the scene changes meaningfully (object pickup, contact events, viewpoint shifts). We score each frame by the smoothed discrete derivative of CLIP image embeddings,

$$d_t \;=\; \big\|\Phi_{\text{CLIP}}(I_t) - \Phi_{\text{CLIP}}(I_{t-1})\big\|_2, \qquad \tilde{d}_t \;=\; (k_{\text{rbf}} * d)_t, \tag{2}$$

where k_{rbf} is a small RBF kernel that suppresses frame-rate jitter. Keyframes are local maxima of $\tilde{d}_t$ exceeding a threshold τ, with the first frame $t = 0$ always retained as an anchor. The detector is purely deterministic preprocessing and has no learnable parameters; τ is a single hyperparameter chosen so that the average inter-keyframe gap is ~ 32 frames.

Keyframe TokenDrop. Storing the full S-token grid for every keyframe is wasteful: most of the image (background, robot base, table) repeats across keyframes. For every keyframe k_i with $i > 0$ we retain only the patches that actually changed relative to its *previous* keyframe k_{i-1}. Concretely, on the 16×16 patch grid we mark a patch as *significant* if its mean per-pixel absolute difference between I_{k_i} and $I_{k_{i-1}}$ exceeds δ, and we keep the union of all significant patches together with the patches lying inside their axis-aligned bounding rectangle (the latter preserves spatial contiguity for downstream attention). The first keyframe k_0 is always kept in full (S tokens); each subsequent keyframe contributes $|\mathcal{P}_i| \le S$ tokens. TokenDrop is also parameter-free.

Motion tokens (MotionJEPA encoding). Independently of keyframe placement, we tile the video with non-overlapping windows of fixed stride Δ (we use $\Delta = 8$). For each window $[t, t+\Delta]$ a trainable *MotionEncoder* f_θ compresses the visual change into M latent tokens $m_t \in \mathbb{R}^{M\times D_m}$:

$$m_t \;=\; f_\theta(\, x_t,\; x_{t+\Delta},\; \Delta s_t \,), \qquad \Delta s_t = s_{t+\Delta} - s_t. \tag{3}$$

Internally, f_θ (i) computes the patchwise visual delta $\Delta x = x_{t+\Delta} - x_t \in \mathbb{R}^{S\times E}$, (ii) spatially average-pools Δx to a single global delta token in $\mathbb{R}^{1\times E}$ and projects it to the encoder width H, (iii) projects Δs_t to H and adds it to the pooled delta token (this is the only place proprioception enters the memory pipeline), (iv) appends M learned read-out queries at the tail, and (v) runs a small causal transformer (RMSNorm, 1D-RoPE causal self-attention, SwiGLU FFN) so that each query attends to the fused delta token before it. The last M positions are projected to motion dimension D_m and constitute m_t.

Memory sequence assembly. The final memory for an episode is the time-ordered concatenation

$$\mathcal{M} \;=\; \big[\, x_{k_0}^{\text{full}},\; m_0,\; m_\Delta,\; \ldots,\; x_{k_i}^{\text{TD}},\; \ldots \,\big], \tag{4}$$

where each entry carries its source timestamp as a positional tag. Compared to the naive grid of $T \cdot S$ patch tokens this yields a roughly order-of-magnitude reduction in token count while preserving both long-range scene anchors and short-range dynamics (concrete ratios are reported in Sec. 3.5).

Table 1: TODO: main results. Brief description plus the key take-away message.

Method	Metric A	Metric B
Baseline	TODO	TODO
Ours	TODO	TODO

Training objective. Keyframe selection and TokenDrop are training-free, so the only learnable parameters in the memory pipeline are those of the MotionEncoder f_θ. We train f_θ jointly with an auxiliary *MotionDiT* decoder g_ϕ under a JEPA-style future-prediction objective on the SigLIP token space. Given $(x_t, x_{t+\Delta})$, we sample a diffusion-style timestep $k \sim \mathcal{U}\{1, \ldots, K\}$ and ask g_ϕ to predict the (token-normalized) future tokens from the current tokens, the motion code, and k:

$$\hat{x}^{(k)}_{t+\Delta} \;=\; g_\phi(\, x_t,\; m_t,\; k\,), \qquad m_t = f_\theta(x_t, x_{t+\Delta}, \Delta s_t), \tag{5}$$

$$\mathcal{L}_{\text{pred}} \;=\; \mathbb{E}_{t,\,k}\Big[\, \big\| \hat{x}^{(k)}_{t+\Delta} \;-\; \mathrm{TokenNorm}(x_{t+\Delta}) \big\|_2^2 \,\Big]. \tag{6}$$

$\mathrm{TokenNorm}$ is a per-feature standardization applied to the target SigLIP tokens to stabilize the regression target. g_ϕ is a small DiT with AdaLN-zero timestep modulation and a tied gated input/output projection initialized to zero, so at $k = 0$ the decoder acts as a normalized identity copy of x_t and only learns residual corrections driven by m_t. This zero-initialized residual head is the sole reason the loss is stable when f_θ is randomly initialized: gradients flow through g_ϕ into f_θ only via the correction term, forcing m_t to encode exactly the information needed to bridge $x_t \to x_{t+\Delta}$.

Crucially, g_ϕ exists only during pre-training. At deployment the robot has already executed both endpoints of every motion window, so memory tokens are produced by a single forward pass through Φ_{SigLIP} and f_θ; no decoding is performed and g_ϕ is discarded. The trained f_θ, together with the parameter-free keyframe and TokenDrop modules, is what ultimately writes into $\mathcal{M}$.

3.4 Training Objective

3.5 Implementation Details

4 Experiments

4.1 Experimental Setup

4.2 Main Results

4.3 Ablation Studies

4.4 Analysis

5 Conclusion

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[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

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